

Note 4.8 Integration by Substitution

Idea

- If $F'(u) = f(u)$, $u = g(x)$ is differentiable, then

$$dF[g(x)] = f[g(x)]g'(x)dx$$

thus

$$\int f[g(x)]g'(x) dx = \int f(u) du = F(u) + C$$

The First Type of Substitution

- **Strategy:**

If $f(x)$ is continuous, $g(x)$ is differentiable, then

$$\int f[g(x)]g'(x) dx = \int f(u) du = \int f[g(x)] dg(x)$$